

VQE LiH Using Parity Mapping

January 6, 2025

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[1]: import matplotlib.pyplot as plt
import numpy as np#second_q.

from qiskit.algorithms import VQE
from qiskit_nature.algorithms import
    ↳(GroundStateEigensolver, NumPyMinimumEigensolverFactory)
from qiskit_nature.drivers import Molecule
from qiskit_nature.drivers.second_quantization import
    ↳(ElectronicStructureMoleculeDriver, ElectronicStructureDriverType)
from qiskit_nature.transformers.second_quantization.electronic import
    ↳FreezeCoreTransformer
from qiskit_nature.problems.second_quantization import
    ↳ElectronicStructureProblem
from qiskit_nature.converters.second_quantization import QubitConverter
from qiskit_nature.mappers.second_quantization import ParityMapper

from qiskit_nature.circuit.library import UCCSD, HartreeFock
from qiskit.circuit.library import EfficientSU2
from qiskit.algorithms.optimizers import COBYLA, SPSA, SLSQP
from qiskit.opflow import TwoQubitReduction
from qiskit import BasicAer, Aer
from qiskit.utils import QuantumInstance
from qiskit.utils.mitigation import CompleteMeasFitter
from qiskit.providers.aer.noise import NoiseModel
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[12]: pip install 'qiskit[nature]'
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Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: qiskit[nature] in
/home/saptarshi/.local/lib/python3.9/site-packages (0.36.2)
Requirement already satisfied: qiskit-ignis==0.7.1 in
/home/saptarshi/.local/lib/python3.9/site-packages (from qiskit[nature]) (0.7.1)
Requirement already satisfied: qiskit-ibmq-provider==0.19.1 in
/home/saptarshi/.local/lib/python3.9/site-packages (from qiskit[nature])
(0.19.1)
Requirement already satisfied: qiskit-terra==0.20.2 in
/home/saptarshi/.local/lib/python3.9/site-packages (from qiskit[nature])
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/home/saptarshi/.local/lib/python3.9/site-packages (from qiskit[nature]) (0.5.2)
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Requirement already satisfied: python-dateutil>=2.8.0 in
/home/saptarshi/.local/lib/python3.9/site-packages (from qiskit-ibmq-
provider==0.19.1->qiskit[nature]) (2.8.2)
Requirement already satisfied: networkx>=0.8.0 in
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ignis==0.7.1->qiskit[nature]) (0.11.0)
Requirement already satisfied: setuptools>=40.1.0 in /usr/lib/python3/dist-
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Requirement already satisfied: python-constraint>=1.4 in
/home/saptarshi/.local/lib/python3.9/site-packages (from qiskit-
terra==0.20.2->qiskit[nature]) (1.4.0)
Requirement already satisfied: ply>=3.10 in /usr/lib/python3/dist-packages (from
qiskit-terra==0.20.2->qiskit[nature]) (3.11)
Requirement already satisfied: symengine>=0.9 in
/home/saptarshi/.local/lib/python3.9/site-packages (from qiskit-
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Requirement already satisfied: stevedore>=3.0.0 in
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Requirement already satisfied: tweedledum<2.0,>=1.1 in
/home/saptarshi/.local/lib/python3.9/site-packages (from qiskit-

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terra==0.20.2->qiskit[nature]) (1.1.1)
Requirement already satisfied: sympy>=1.3 in /usr/lib/python3/dist-packages
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nature>=0.3.0->qiskit[nature]) (3.8.0)
Requirement already satisfied: rustworkx in
/home/saptarshi/.local/lib/python3.9/site-packages (from qiskit-
nature>=0.3.0->qiskit[nature]) (0.12.1)
Requirement already satisfied: typing-extensions in /usr/lib/python3/dist-
packages (from qiskit-nature>=0.3.0->qiskit[nature]) (3.10.0.2)
Collecting qiskit-nature>=0.3.0
  Downloading qiskit_nature-0.5.1-py3-none-any.whl (3.8 MB)

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nature>=0.3.0->qiskit[nature]) (1.2.2)
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Requirement already satisfied: six>=1.5 in /usr/lib/python3/dist-packages
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(1.16.0)
Requirement already satisfied: joblib>=1.1.1 in
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learn>=0.20.0->qiskit-nature>=0.3.0->qiskit[nature]) (1.2.0)
Requirement already satisfied: threadpoolctl>=2.0.0 in
/home/saptarshi/.local/lib/python3.9/site-packages (from scikit-
learn>=0.20.0->qiskit-nature>=0.3.0->qiskit[nature]) (3.1.0)
Requirement already satisfied: pbr!=2.1.0,>=2.0.0 in
/home/saptarshi/.local/lib/python3.9/site-packages (from
stevedore>=3.0.0->qiskit-terra==0.20.2->qiskit[nature]) (5.9.0)
Installing collected packages: qiskit-nature
  Attempting uninstall: qiskit-nature
    Found existing installation: qiskit-nature 0.5.2
    Uninstalling qiskit-nature-0.5.2:
      Successfully uninstalled qiskit-nature-0.5.2
Successfully installed qiskit-nature-0.4.1
WARNING: There was an error checking the latest version of pip.

Note: you may need to restart the kernel to use updated packages.

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```

[2]: def get_qubit_op(dist):
    # Define Molecule
    molecule = Molecule(
        # Coordinates in Angstrom
        geometry=[
            ["Li", [0.0, 0.0, 0.0]],
            ["H", [dist, 0.0, 0.0]]
        ],
        multiplicity=1, # = 2*spin + 1
        charge=0,
    )

    driver = ElectronicStructureMoleculeDriver(
        molecule=molecule,
        basis="sto3g",
        driver_type=ElectronicStructureDriverType.PYSCF)

    # Get properties
    properties = driver.run()
    num_particles = (properties
                     .get_property("ParticleNumber")
                     .num_particles)
    num_spin_orbitals = int(properties
                             .get_property("ParticleNumber")
                             .num_spin_orbitals)

    # Define Problem, Use freeze core approximation, remove orbitals.
    problem = ElectronicStructureProblem(
        driver,
        [FreezeCoreTransformer(freeze_core=True,
                              remove_orbitals=[-3,-2])])

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second_q_ops = problem.second_q_ops() # Get 2nd Quant OP
num_spin_orbitals = problem.num_spin_orbitals
num_particles = problem.num_particles

mapper = ParityMapper() # Set Mapper
hamiltonian = second_q_ops[0] # Set Hamiltonian
# Do two qubit reduction
converter = QubitConverter(mapper,two_qubit_reduction=True)
reducer = TwoQubitReduction(num_particles)
qubit_op = converter.convert(hamiltonian)
qubit_op = reducer.convert(qubit_op)

return qubit_op, num_particles, num_spin_orbitals, problem, converter

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[4]: def exact_solver(problem , converter):
    solver = NumPyMinimumEigensolverFactory()
    calc = GroundStateEigensolver(converter,solver) #converter,
    result = calc.solve(problem)
    return result

backend = BasicAer.get_backend("statevector_simulator")
distances = np.arange(0.5, 4.0, 0.1)
exact_energies = []
vqe_energies = []
optimizer = SLSQP(maxiter=5)

# pylint: disable=undefined-loop-variable
for dist in distances:
    (qubit_op, num_particles, num_spin_orbitals,problem, converter) = get_qubit_op(dist)
    result = exact_solver(problem,converter)
    exact_energies.append(result.total_energies[0].real)
    init_state = HartreeFock(num_spin_orbitals, num_particles, converter)
    var_form = UCCSD(converter,num_particles,num_spin_orbitals,initial_state=init_state)#converter,
    vqe = VQE(var_form, optimizer, quantum_instance=backend)
    vqe_calc = vqe.compute_minimum_eigenvalue(qubit_op)
    vqe_result = problem.interpret(vqe_calc).total_energies[0].real
    vqe_energies.append(vqe_result)
    print(f"Interatomic Distance: {np.round(dist, 2)}",
          f"VQE Result: {vqe_result:.5f}",
          f"Exact Energy: {exact_energies[-1]:.5f}")

print("All energies have been calculated")

```

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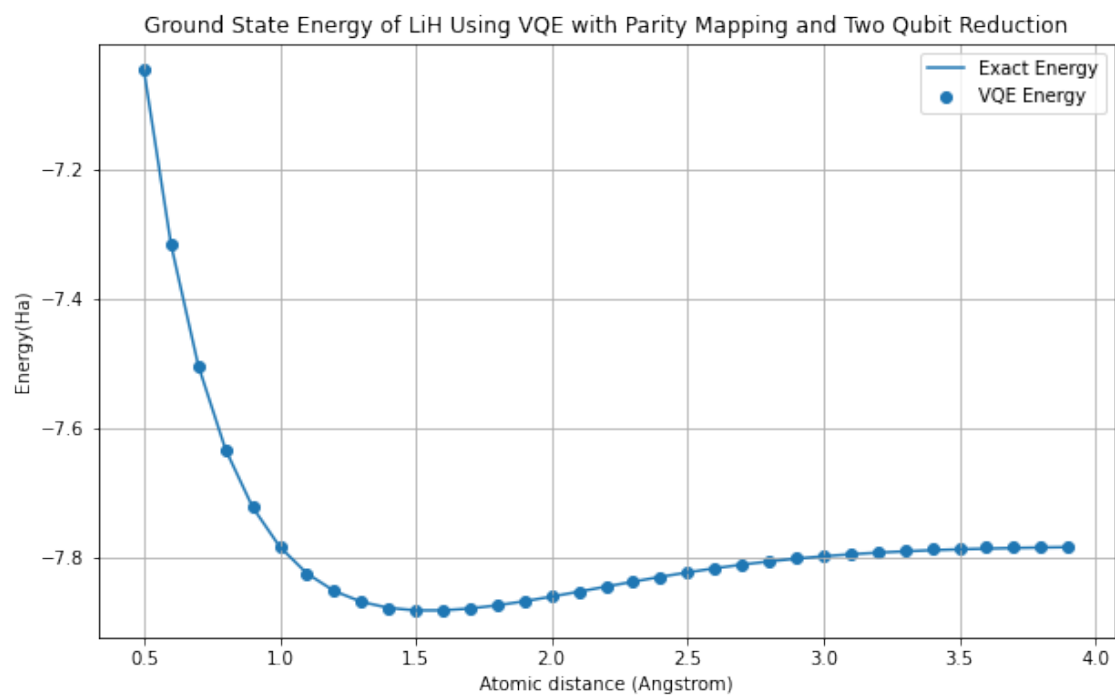
Interatomic Distance: 0.5 VQE Result: -7.04751 Exact Energy: -7.04791
Interatomic Distance: 0.6 VQE Result: -7.31769 Exact Energy: -7.31784

```

Interatomic Distance: 0.7 VQE Result: -7.50392 Exact Energy: -7.50400
 Interatomic Distance: 0.8 VQE Result: -7.63332 Exact Energy: -7.63338
 Interatomic Distance: 0.9 VQE Result: -7.72278 Exact Energy: -7.72283
 Interatomic Distance: 1.0 VQE Result: -7.78398 Exact Energy: -7.78402
 Interatomic Distance: 1.1 VQE Result: -7.82516 Exact Energy: -7.82520
 Interatomic Distance: 1.2 VQE Result: -7.85212 Exact Energy: -7.85216
 Interatomic Distance: 1.3 VQE Result: -7.86886 Exact Energy: -7.86890
 Interatomic Distance: 1.4 VQE Result: -7.87819 Exact Energy: -7.87823
 Interatomic Distance: 1.5 VQE Result: -7.88210 Exact Energy: -7.88214
 Interatomic Distance: 1.6 VQE Result: -7.88206 Exact Energy: -7.88210
 Interatomic Distance: 1.7 VQE Result: -7.87917 Exact Energy: -7.87920
 Interatomic Distance: 1.8 VQE Result: -7.87417 Exact Energy: -7.87428
 Interatomic Distance: 1.9 VQE Result: -7.86788 Exact Energy: -7.86799
 Interatomic Distance: 2.0 VQE Result: -7.86073 Exact Energy: -7.86083
 Interatomic Distance: 2.1 VQE Result: -7.85312 Exact Energy: -7.85320
 Interatomic Distance: 2.2 VQE Result: -7.84534 Exact Energy: -7.84541
 Interatomic Distance: 2.3 VQE Result: -7.83763 Exact Energy: -7.83772
 Interatomic Distance: 2.4 VQE Result: -7.83021 Exact Energy: -7.83034
 Interatomic Distance: 2.5 VQE Result: -7.82324 Exact Energy: -7.82343
 Interatomic Distance: 2.6 VQE Result: -7.81685 Exact Energy: -7.81709
 Interatomic Distance: 2.7 VQE Result: -7.81116 Exact Energy: -7.81142
 Interatomic Distance: 2.8 VQE Result: -7.80620 Exact Energy: -7.80644
 Interatomic Distance: 2.9 VQE Result: -7.80195 Exact Energy: -7.80215
 Interatomic Distance: 3.0 VQE Result: -7.79832 Exact Energy: -7.79850
 Interatomic Distance: 3.1 VQE Result: -7.79516 Exact Energy: -7.79545
 Interatomic Distance: 3.2 VQE Result: -7.79233 Exact Energy: -7.79293
 Interatomic Distance: 3.3 VQE Result: -7.78973 Exact Energy: -7.79085
 Interatomic Distance: 3.4 VQE Result: -7.78748 Exact Energy: -7.78915
 Interatomic Distance: 3.5 VQE Result: -7.78572 Exact Energy: -7.78776
 Interatomic Distance: 3.6 VQE Result: -7.78443 Exact Energy: -7.78664
 Interatomic Distance: 3.7 VQE Result: -7.78351 Exact Energy: -7.78573
 Interatomic Distance: 3.8 VQE Result: -7.78288 Exact Energy: -7.78500
 Interatomic Distance: 3.9 VQE Result: -7.78245 Exact Energy: -7.78441
 All energies have been calculated

```

[9]: plt.figure(1,figsize=(10,6))
plt.plot(distances, exact_energies, label="Exact Energy")
plt.scatter(distances, vqe_energies, label="VQE Energy")
plt.title("Ground State Energy of LiH Using VQE with Parity Mapping and Two-
    ↪Qubit Reduction")
plt.xlabel('Atomic distance (Angstrom)')
plt.ylabel('Energy(Ha)')
plt.legend()
plt.grid()
plt.savefig('VQE2.png',dpi=500)
  
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